

Decomposition-Based Anomaly Detection, RPCA, Robust GAN Inversion, and Beyond

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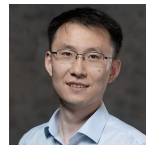
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Outline

1. RPCA and decomposition-based anomaly detection (10 min)
2. From matrices to tensors, and to resource-limited settings (15 min)
 - ▶ Additive Tensor Decomposition (ATD)
 - ▶ Compressed Smooth Sparse Decomposition (CSSD)
3. Beyond linear priors: Robust GAN Inversion (15 min)
 - ▶ RGI / R-RGI (images)
 - ▶ Uni-3DAD (3D point clouds)
4. The VISION Workshop @ CVPR 2026 (5 min)

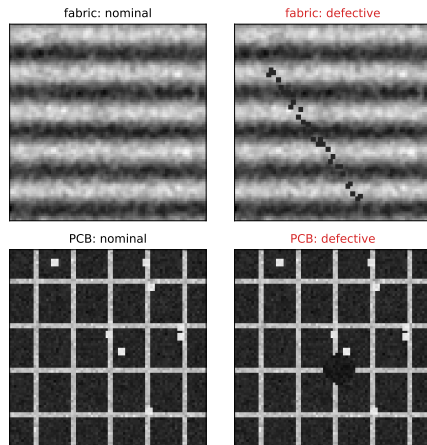
Motivation: industrial defect detection

Image-based inspection is a core workhorse in manufacturing QC.

A single product image typically decomposes into two parts:

- ▶ a **background** that is shared across nominal products (texture, layout, illumination), and
- ▶ a **defect** that is *rare* and *localized*.

Question: can we separate these two *without labels*, purely from structural priors?



From a stack of images to a data matrix

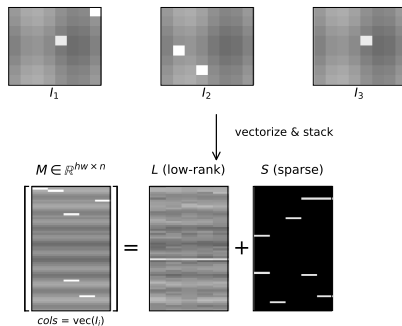
Collect n images, each of size $h \times w$.

1. Vectorize each image: $\text{vec}(I_i) \in \mathbb{R}^{hw}$.
2. Stack as columns of a matrix:

$$M = [\text{vec}(I_1) \text{vec}(I_2) \cdots \text{vec}(I_n)] \in \mathbb{R}^{hw \times n}.$$

Key observations:

- ▶ Common background across images $\Rightarrow$ the background component has **low rank**.
- ▶ Defects are isolated and differ image to image $\Rightarrow$ the defect component is **sparse**.



Vectorize each image into a column; stack to form M ;
then $M = L + S$.

Robust PCA (RPCA)

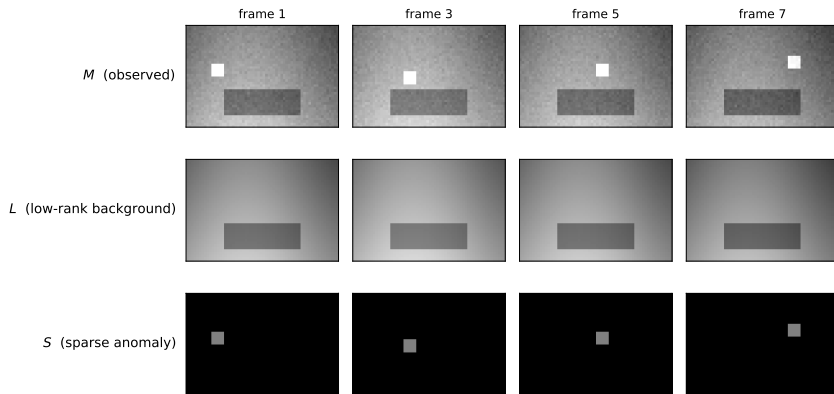
Candès, Li, Ma, Wright (2011) — recover L and S from M :

$$\min_{L, S} \underbrace{\|L\|_*}_{\text{low-rank prior}} + \lambda \underbrace{\|S\|_1}_{\text{sparse prior}} \quad \text{s.t.} \quad M = L + S.$$

- ▶ $\|\cdot\|_*$: nuclear norm (sum of singular values) — convex surrogate of $\text{rank}(\cdot)$.
- ▶ $\|\cdot\|_1$: entrywise ℓ_1 — convex surrogate of $\|\cdot\|_0$.
- ▶ Provable exact recovery under incoherence / sparsity conditions.
- ▶ Solved efficiently by ADMM / ALM; mature toolbox.

For inspection: columns of L are denoised nominal images; columns of S are pixel-level anomaly maps.

RPCA as an anomaly detector — the picture



RPCA works well when: background is *truly low rank*, anomalies are *sparse* and *small in magnitude*, and noise is mild.

Where plain RPCA struggles

- ▶ **Tensor-structured data.** CT / video / multi-channel imaging has structure along several modes simultaneously (space *and* time, space *and* spectrum). Reshaping to a matrix throws that structure away.
- ▶ **Resource-limited acquisition.** High-resolution sensors are fast and produce huge data; storing, transmitting, and inverting M becomes a bottleneck.
- ▶ **Complex, non-linear backgrounds.** Textured products (wafers, fabrics, leather) live on a manifold, not a low-rank subspace. Low-rank priors are too crude.
- ▶ **Streaming / online / mixed datasets.** Nominal appearance drifts; products change.

Rest of this talk: how to generalize decomposition-based anomaly detection along each of these axes.

Part 2 — Generalizing RPCA

1. Additive Tensor Decomposition (ATD)

Mou, Wang, Zhang, Shi. *IEEE Trans. Autom. Sci. Eng.*, 2021.

Decomposition that respects tensor structure (smoothness, sparsity, low-rank, piecewise-constancy across individual modes).

2. Compressed Smooth Sparse Decomposition (CSSD)

Mou, Shi. *INFORMS Journal on Data Science*, 2023.

One-step acquisition + decomposition; fast and data-efficient for resource-limited inspection.

Why tensors? A medical-imaging motivating example

CT scan of the aortic root: a *stack* of images along the scanning direction.

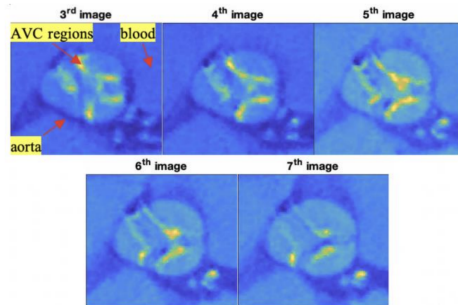
Tensor $\mathcal{M} \in \mathbb{R}^{l_1 \times l_2 \times l_3}$:

- ▶ mode 1: slice index (spatial depth),
- ▶ modes 2–3: in-slice pixels.

We want three components:

- ▶ background (blood pool + soft tissue),
- ▶ **AVC regions** (aortic valve calcification — the clinical target),
- ▶ measurement error.

Exact same story applies to defects on rolled steel, thermal hotspots, X-ray CT of parts, ...

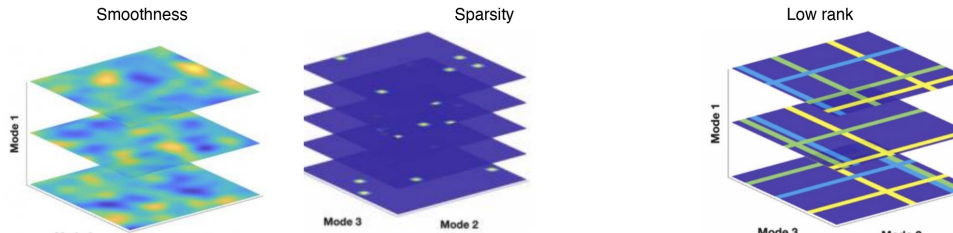


CT slices of the aortic root: bright calcifications attach to the inner wall, vary across slices.

Additive Tensor Decomposition (ATD) — framework

Decompose $\mathcal{M} = \sum_{i=1}^m \mathcal{X}_i$ with *distinct* structural priors on each component $\mathcal{X}_i$:

$$\min_{\mathcal{X}_1, \dots, \mathcal{X}_m} \sum_{i=1}^m \sum_{j=1}^{n_i} \lambda_{i,j} p_{i,j}(\mathcal{X}_i) \quad \text{s.t.} \quad \mathcal{M} = \sum_{i=1}^m \mathcal{X}_i.$$



Structural priors $p_{i,j}$ are defined *per mode*:

- ▶ **Smoothness:** $\sum_s \|D_s \mathcal{X}_{(s)}\|_F^2$
- ▶ **Sparsity:** entrywise ℓ_1 or group lasso

- ▶ **Low rank:** $\|\mathcal{X}_{(j)}\|_*$
- ▶ **Piecewise constancy:** fused lasso

RPCA, SSD, ST-SSD are *special cases* of ATD.

ATD — algorithm and why it scales

Solved by the **ADMM** algorithm:

- ▶ The separable structure of f and g lets the proximal operator decompose across components $\mathcal{X}_i$; each $p_{i,j}$ admits a closed-form proximal update (paper, Sec. III-B).
- ▶ Step (2) (updating f): update of each tensor component can be done fiber by fiber.
- ▶ Step (3) (updating g): the element update reduces to a small linear system; the projection matrix $\mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^{-1}$ is computed *once*.
- ▶ $\Rightarrow$ “highly parallelizable, suitable for high-dimensional data analysis.”

Algorithm 1 ADMM algorithm

Initialize $\tilde{\mathcal{Z}}$ and $\tilde{\mathcal{U}}$ as the same data structure as $\tilde{\mathcal{X}}$, with all their elements being 0.

Do:

- (1) Save $(\tilde{\mathcal{Z}}_{\text{prev}}, \tilde{\mathcal{U}}_{\text{prev}}) \leftarrow (\tilde{\mathcal{Z}}, \tilde{\mathcal{U}})$.
- (2) Update $\tilde{\mathcal{X}} = \text{prox}_{\eta f}(\tilde{\mathcal{Z}}_{\text{prev}} - \tilde{\mathcal{U}}_{\text{prev}})$.
- (3) Update $\tilde{\mathcal{Z}} = \text{prox}_{\eta g}(\tilde{\mathcal{X}} - \tilde{\mathcal{U}}_{\text{prev}})$.
- (4) Update $\tilde{\mathcal{U}} = \tilde{\mathcal{U}}_{\text{prev}} + \tilde{\mathcal{X}} - \tilde{\mathcal{Z}}$.

Until: $\|\tilde{\mathcal{U}} - \tilde{\mathcal{U}}_{\text{prev}}\| < \epsilon, \|\tilde{\mathcal{Z}} - \tilde{\mathcal{Z}}_{\text{prev}}\| < \epsilon$.

ADMM outer loop (Mou et al., 2021).

ATD — results

Two simulations + a real CT case study:

- **Crack growth** (irregular illumination): SSD captures only some pixels on the crack — its ℓ_1 penalty shrinks low-intensity crack pixels to zero. ATD exploits temporal smoothness and preserves those pixels.
- **Thermal hotspots** (static + moving on smooth background): RPCA identifies the moving object but *cannot* separate the static hotspot from the background; ATD separates background, static hotspot, and moving hotspot.
- **AVC regions in CT**: ATD decomposes the scan into background (blood pool + soft tissue), AVC regions, and measurement error by exploiting slice-wise smoothness and sparsity.

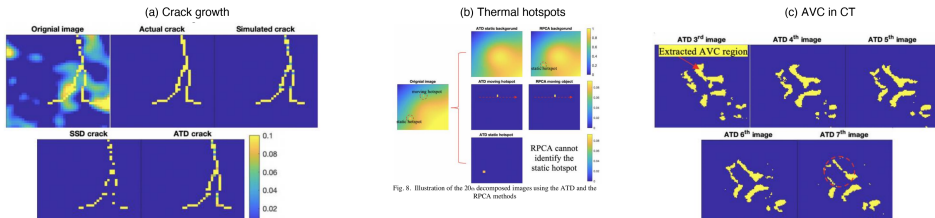


Fig. 8. Illustration of the 20x-decomposed images using the ATD and the RPCA methods

Figures from Mou et al., 2021. Takeaway: *per-mode* structural priors matter.

Resource-limited setting: the new bottleneck

Modern inspection cameras are fast *and* high-resolution.

- ▶ Acquisition rates keep climbing.
- ▶ But: storage, transmission, and decomposition over full images become the bottleneck — not the physics.

Question. Can we *skip* acquiring the full image and still recover the anomaly?

CSSD (Mou & Shi, 2023): fuse *compressive acquisition* with *smooth + sparse decomposition* in a single optimization.

Smooth + Sparse Decomposition (SSD), briefly

Yan, Paynabar & Shi, *Technometrics*, 2017.

For a single image y , model $y = B\theta_B + A\theta_A + \varepsilon$, where

- ▶ B : smooth basis (e.g. 2D B-splines) — background.
- ▶ A : identity / indicator basis — per-pixel anomalies.

Estimate by

$$\min_{\theta_B, \theta_A} \frac{1}{2} \|y - B\theta_B - A\theta_A\|_2^2 + \lambda_B R_B(\theta_B) + \lambda_A \|\theta_A\|_1,$$

where $R_B(\theta_B) = \|D\theta_B\|_2^2$ is a roughness penalty (D is a finite-difference operator) that enforces smoothness of the background. Nice for inspection: interpretable, fast, theoretical guarantees.

But: still requires the full image y as input.

Compressed Smooth Sparse Decomposition (CSSD)

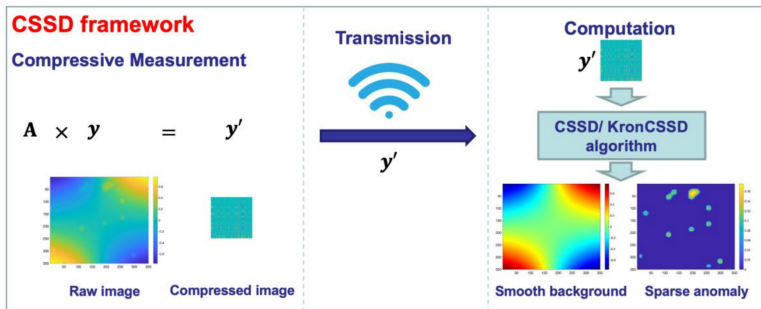
Idea. Replace y by compressive measurements $z = \Phi y$, $\Phi \in \mathbb{R}^{m \times p}$ with $m \ll p$:

$$\min_{\theta_B, \theta_A} \frac{1}{2} \|z - \Phi(B\theta_B + A\theta_A)\|_2^2 + \lambda_B R_B(\theta_B) + \lambda_A \|\theta_A\|_1.$$

- ▶ **One-step:** no reconstruct-then-decompose cascade.
- ▶ **Provable guarantees** on support recovery of the anomaly component under RIP-like conditions on Φ restricted to the joint dictionary $[\Phi B, \Phi A]$.
- ▶ **KronCSSD:** structured $\Phi = \Phi_1 \otimes \Phi_2$ for high-dim images — huge memory & speed win, tiny accuracy loss.

CSSD — pipeline at a glance

Figure 1. (Color online) An Overview of the CSSD/KronCSSD Framework



Acquisition and decomposition are performed *jointly*: no reconstruct-then-decompose cascade. $m \ll p$ measurements suffice when the background is low-dimensional in B and the anomaly is sparse in pixel space.

(Figure from Mou & Shi, 2023.)

CSSD — case studies

Three applications: (a) raw image, (b) KronCSSD result, (c) SSD result.

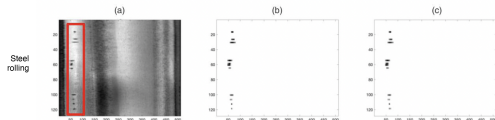
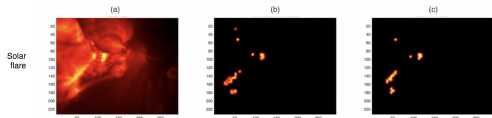
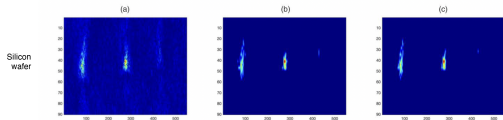


Figure 11. (Color online) Solar Activity Images and Detected Solar Flare



Notes. (a) Raw image. (b) KronCSSD result. (c) SSD result.

Figure 12. (Color online) Silicon Stress Map and Detected Indentation



Notes. (a) Raw image. (b) KronCSSD result. (c) SSD result.

Figures from Mou & Shi, 2023 — KronCSSD matches SSD with a fraction of the measurements.

Part 2 summary

- ▶ **ATD** lifts the matrix story to tensors by decoupling structural priors across modes: smoothness, sparsity, low-rank, and piecewise-constancy, each applied where it belongs.
- ▶ **CSSD** lifts the pipeline to resource-limited regimes: acquisition and decomposition are performed *jointly*, with provable anomaly recovery under far fewer measurements.

Still missing: what if the background isn't low-rank, smooth, or structurally simple — but lives on a *learned, non-linear* manifold?

⇒ replace the linear prior with a *deep generative* prior.

Part 3 — Beyond linear priors

1. RGI / R-RGI

Mou, Gu, Cao, Bai, Huang, Shan, Shi. *ICLR 2023*.

Decomposition with a deep generative prior — robust GAN inversion for unsupervised pixel-wise anomaly detection and mask-free inpainting.

2. Uni-3DAD

Liu, Mou, Gaw, Wang. *Expert Systems with Applications*, 2025.

Carrying the idea to 3D point clouds: universal anomaly detection on model-free products.

When low-rank priors fail

Textured products (wafers, fabrics, LCD screens) have **complex, non-low-rank** backgrounds.

A GAN trained on nominal products is a good approximator of the *natural image manifold*:

$$\mathcal{G} : \mathbb{R}^d \rightarrow \mathbb{R}^{h \times w}, \quad \text{“normal” images} \approx \text{range}(\mathcal{G}).$$

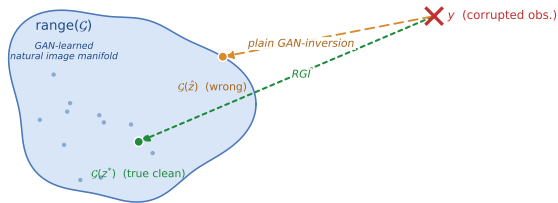
GAN inversion (Zhu et al., 2016). Given an image y , solve

$$\hat{z} = \arg \min_z L_{\text{rec}}(y, \mathcal{G}(z)).$$

For *clean* inputs this projects y onto $\text{range}(\mathcal{G})$, and $y - \mathcal{G}(\hat{z})$ highlights what is not normal.

Failure mode. Under *gross, unknown* corruptions s^* with $\|s^*\|_0 \leq n_0$, plain inversion projects y onto the *nearest boundary point* of $\text{range}(\mathcal{G})$ — the recovered $\mathcal{G}(\hat{z})$ drifts *toward* the corruption, and the residual is a poor anomaly mask.

From “GAN inversion” to “robust GAN inversion”



- **plain inversion** projects y to the nearest point of $\text{range}(\mathcal{G})$;
- **RGI** peels off a sparse s and lands at the true $\mathcal{G}(z^*)$.

Decomposition view

$$y = \mathcal{G}(z) + s + \varepsilon,$$

with $\mathcal{G}(z) \in \text{range}(\mathcal{G})$ and s *sparse*.

RGI objective

$$\min_{z, s} \frac{1}{2} \|y - \mathcal{G}(z) - s\|_2^2 + \lambda \|s\|_1.$$

- ▶ *Non-linear* low-complexity term $\mathcal{G}(z)$ replaces the nuclear norm.
- ▶ Sparse residual s absorbs the gross corruption.
- ▶ Same spirit as RPCA – “normal + rare”.

Robust GAN Inversion (RGI)

Model. Assume $x = \mathcal{G}(z^*) + s^*$ with $\|s^*\|_0 \leq n_0$, and let $M^* = \mathbf{1}_{\{s^* \neq 0\}}$. Learn the latent z and a binary mask M jointly, ignoring the corrupted pixels during reconstruction:

$$\min_{z \in \mathbb{R}^d, M \in \{0,1\}^{m \times n}} L_{\text{rec}}((\mathbf{1} - M) \odot x, (\mathbf{1} - M) \odot \mathcal{G}(z)) \quad \text{s.t.} \quad \|M\|_0 \leq n_0. \quad (\text{Eq. 2})$$

Relaxation (RGI). Drop the hard constraint; penalise mask size:

$$\min_{z, M} L_{\text{rec}}((\mathbf{1} - M) \odot x, (\mathbf{1} - M) \odot \mathcal{G}(z)) + \lambda \|M\|_1, \quad \text{solved by gradient descent on } (z, M). \quad (\text{Eq. 3})$$

Guarantees (Mou et al., ICLR 2023):

- ▶ **Theorem 1 (image).** $d_{\infty}(\hat{z}(\lambda), \mathcal{Z}^*) \downarrow 0$ as $\lambda \downarrow 0$, hence $\mathcal{G}(\hat{z}) \rightarrow \mathcal{G}(z^*)$ — the restored image converges to the true clean background regardless of corruption magnitude.
- ▶ **Theorem 2 (mask).** $\hat{M}(\lambda) \rightarrow M^*$ as $\lambda \downarrow 0$; there is a finite $\bar{\lambda} > 0$ such that for any $\lambda \leq \bar{\lambda}$, M^* is obtained by simply *thresholding* $\hat{M}$.
- ▶ Same decomposition philosophy as RPCA: nonlinear low-complexity term $\mathcal{G}(z)$ replaces the nuclear norm; $\|M\|_1$ replaces $\|S\|_1$.

R-RGI — closing the GAN approximation gap

Real nominal images don't lie *exactly* on $\text{range}(\mathcal{G})$: a gap $\|x - \mathcal{G}(z^*)\|$ persists even without corruption. Yet blind generator fine-tuning trivially overfits the corrupted input.

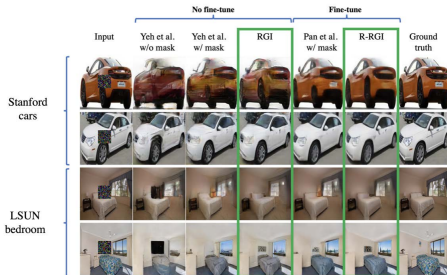
R-RGI (Relaxed RGI). Make θ a decision variable too:

$$\min_{z \in \mathbb{R}^d, M \in \{0,1\}^{m \times n}, \theta \in \mathbb{R}^w} L_{\text{rec}}((\mathbf{1} - M) \odot x, (\mathbf{1} - M) \odot \mathcal{G}(z; \theta)) + \lambda \|M\|_1. \quad (\text{Eq. 4})$$

Training schedule. Freeze θ early $\rightarrow$ once (z, M) stabilise, jointly optimise θ with a small step size for a limited number of iterations.

- ▶ The $(\mathbf{1} - M) \odot$ factor means fine-tuning only uses *uncorrupted* pixels – the generator cannot memorise the defect.
- ▶ Closes the GAN approximation gap $\rightarrow$ sharper restoration and tighter anomaly masks.
- ▶ **One objective, two applications:** (i) mask-free semantic inpainting (output $\mathcal{G}(\hat{z}; \hat{\theta})$); (ii) unsupervised pixel-level anomaly segmentation (thresholded $\hat{M}$).

RGI — (i) mask-free semantic inpainting



Stanford Cars and LSUN bedroom. **RGI** = no fine-tune, no input mask. **R-RGI** = with generator fine-tune.

No fine-tune (PSNR $\uparrow$)		
Dataset	Yeh'17 w/o mask	RGI
CelebA	11.50	19.70
Cars	16.57	16.89
LSUN bed	16.15	17.67

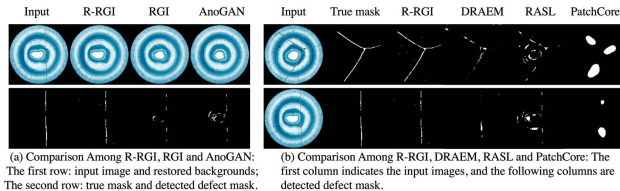
+ generator fine-tune		
Dataset	Pan'21 w/ mask	R-RGI
CelebA	21.74	20.05
Cars	20.98	19.31
LSUN bed	21.36	18.72

Takeaway. RGI is *comparable* to Yeh'17 *w/ mask* (the best no-fine-tune baseline) *without* any pre-configured mask (+8.2 dB on CelebA over the no-mask baseline). R-RGI approaches Pan'21 w/ mask with mask-free fine-tuning.

Tables use Case (i), central-block missing. Figure 3 &

Table 1, Mou et al., ICLR 2023.

RGI — (ii) unsupervised pixel-wise anomaly detection



(a) R-RGI / RGI / AnoGAN: restored backgrounds + defect masks. (b) R-RGI / DRAEM / RASL / PatchCore: defect masks. Synthetic defects (Product03, BTAD).

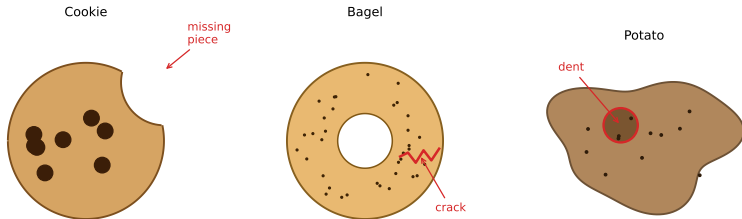
Dice $\uparrow$ on synthetic defects				
method	crack	irregular	scratch	mixed large
RASL	0.293	0.246	0.280	0.355
PatchCore	0.277	0.448	0.350	0.683
AnoGAN	0.457	0.422	0.436	0.540
RGI	0.458	0.420	0.430	0.617
DRAEM	0.473	0.697	0.639	0.825
R-RGI	0.809	0.758	0.745	0.810

Takeaway. R-RGI is best on 3 of 4 defect types (crack, irregular, scratch) by large Dice margins, and within one standard deviation of DRAEM on *mixed large* (0.810 vs 0.825) — *without* any defect supervision, while DRAEM trains on simulated defects.

Figure 4 & Table 2, Mou et al., ICLR 2023.

From pixels to 3D: why point clouds?

Model-free products: no CAD, defects vary by instance



Why 3D in manufacturing:

- ▶ Robust to lighting / texture changes.
- ▶ Captures geometric defects (dents, cracks, *missing pieces*) directly.

Two dominant families — both limited:

- ▶ **CAD-based:** compare scan to design. Fails for “model-free” products (produce, dentures, bones).
- ▶ **Memory-bank:** compare patches to a library of normal patches. Fails on *incomplete* shapes.

Unified, unsupervised 3D anomaly detection on model-free products — two branches + OCSVM fusion:

(1) Feature-based branch.

- ▶ *FPFH* (Fast Point Feature Histogram): manual local descriptor, cheap, no training.
- ▶ *PatchCore* coreset sub-sampling of the memory bank $\mathcal{X}_M \rightarrow \mathcal{M}_c$ (size l) for fast inference.
- ▶ Per-point score:
 $s_1(a_i) = \min_{j \leq l} \|\phi(a_i) - \phi(a_j)\|_2$; output $\mathbf{S}_1$.
- ▶ Catches **local geometric defects**: dents, holes, cracks, scratches.

(2) Reconstruction branch (GAN-Inversion).

- ▶ Train SP-GAN $G(\cdot; \theta)$ on nominal shapes.
- ▶ At test: joint optimisation of latent code *and* generator weights,

$$(\mathbf{z}^*, \theta^*) = \arg \min_{\mathbf{z}, \theta} d_{CD}(G(\mathbf{z}; \theta), P_{in}),$$

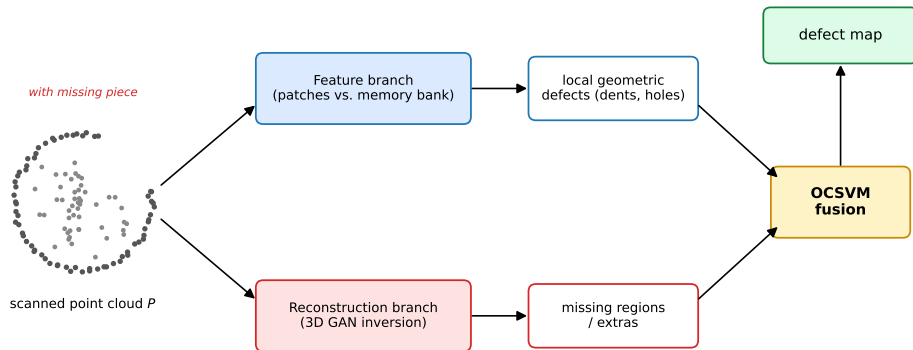
d_{CD} = Chamfer distance.

- ▶ $\mathbf{S}_2 = \|P'_{rec} - P'_{in}\|^2$ (pointwise).
- ▶ Catches **missing regions / extras** that feature methods cannot.

(3) OCSVM fusion. Schölkopf's one-class SVM fuses $(\mathbf{S}_1, \mathbf{S}_2)$ into the final anomaly map — no hand-tuned weights, handles scale mismatch between branches.

Evaluated on MVTec 3D-AD + an augmented incomplete-shape variant; SOTA on incomplete shapes, on-par elsewhere.

Uni-3DAD — the pipeline at a glance



First unified framework that simultaneously handles local geometric defects *and* large missing regions on model-free 3D products.

Part 3 summary — and the common thread

	Low-complexity prior	Sparse prior
RPCA	$\ L\ _*$	$\ S\ _1$
ATD	per-mode low-rank / smoothness	per-mode sparsity
SSD / CSSD	smooth basis $B\theta_B$	$\ \theta_A\ _1$
RGI	deep generator $\mathcal{G}(z)$	$\ M\ _1$ (mask)
Uni-3DAD	3D generator $\mathcal{G}_{3D}(z)$	sparse point set $\mathcal{S}$

Same *decomposition philosophy*: **normal** + **rare**.

The prior on “normal” evolves from *linear subspace* to *tensor-structured* to *deep manifold*.

VISION'26 Workshop @ CVPR 2026 — please join us!

4th Workshop on Vision-based Industrial Inspection

CVPR 2026, Denver, Colorado.

Now incorporates two allied workshops:

- ▶ **DEXTER** — Degraded X-ray Image Tomography, Enhancement, and Reconstruction.
- ▶ **INSIDE-X** — Intelligent Screening and Imaging for Detection and Exploitation via X-ray.

Three tracks: *VISION general*, *DEXTER*, *INSIDE-X*.



Overview Program Call for Papers Organizers Prior Workshops Follow us

CVPR VISION'26
DENVER, COLORADO

GyF

4th workshop on Vision-based Industrial Inspection (VISION)

VISION'26 @ CVPR: Vision-Based Industrial Inspection Across Modalities (Surface, X-ray, 3D & More)

Now incorporating the "DEXTER: Degraded X-ray Image Tomography, Enhancement, and Reconstruction" and "INSIDE-X: Intelligent Screening and Imaging For Detection and Exploitation via X-ray" workshops.

CVPR 2026 Workshops — Denver, Colorado (Date/Room TBA)

Overview

Aims and Scope

The VISION workshop will provide a platform for the exchange of scholarly innovations and emerging practical challenges in Vision-based Industrial Inspection. Through a series of keynote talks, technical presentations, and challenge competitions, this workshop aims to i) bring together researchers from the interdisciplinary research communities related to computer vision-based inspection; and ii) connect researchers and industry practitioners to synergize recent research progress and current needs in industrial practice.

In 2026, VISION expands its scope to reflect the merger with the "DEgraded X-ray Image Tomography, Enhancement, and Reconstruction" and "Intelligent Screening and Imaging For Detection and Exploitation via X-ray" workshops.

We explicitly welcome contributions across these three tracks, from traditional vision based methods to cross-modality methods and shared challenges in real-world inspection pipelines.

Topics include surface imaging, X-ray, CT, and volumetric reconstruction, 3D sensing modalities, computational imaging, imaging physics, multimodal cross-domain fusion, scene understanding for threat detection, datasets and benchmarking, and real-world deployment and applications.

VISION'26 — scope and what we're looking for

Surface imaging, X-ray / CT, 3D sensing, computational imaging, imaging physics, multimodal fusion.

Academic topics:

- ▶ Algorithms for data and annotation limitations.
- ▶ Robust, explainable, accountable ML for inspection.
- ▶ Foundation models in industrial inspection.
- ▶ Multi-modality / multi-energy data fusion.
- ▶ Physics-informed deep learning for material classification.
- ▶ Generative / diffusion models for synthetic X-ray and CT data.

Industry topics:

- ▶ Edge / real-time deployment; non-destructive testing.
- ▶ Automated screening at airports, borders, logistics.
- ▶ Large-scale inspection & screening datasets.

Keynote speakers include Salman Khan (MBZUAI), Dan Cristian Dinca (Raspican), Charles Bouman (Purdue), and more TBA.

Prior editions — and what grew out of them

- ▶ **VISION'23** @ CVPR, Vancouver.
- ▶ **VISION'24** @ ECCV, Milano.
- ▶ **VISION'25** @ ICCV, Honolulu.
- ▶ **VISION'26** @ CVPR, Denver — *now merged with DEXTER & INSIDE-X*.

Data challenge & benchmark each year; a growing community of academia + industry (Apple, Safran, Intel, ...).

Call to action.

- ▶ Submit a paper (regular or extended abstract).
- ▶ Participate in the data challenge.
- ▶ Attend and help shape the field.

<https://vision-workshop-26.github.io/cvpr-2026/>

VISION Datasets — a benchmark for industrial inspection

Bai, Likhomanenko, Tuzel, Huang, Shan, Cao, Mou, Shi, Cinbis (Apple, Georgia Tech, METU), 2023.

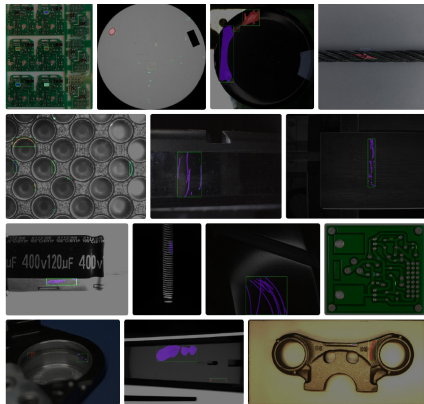


Figure 1: Overview of the VISION Datasets.

Overview of the VISION Datasets (Figure 1 of Bai et al., 2023).

What it is. A curated collection of **14 industrial inspection datasets** (sourced from Roboflow, re-annotated), totalling **18k images** and **44 defect types**.

Scope. Broad coverage of manufacturing processes, materials, and product geometries: PCBs, wafers, fabrics, capacitors, metal parts, wood, ...

Two companion challenge tracks:

- ▶ **Track 1:** data-efficient defect detection.
- ▶ **Track 2:** data-centric, “data → model”.

huggingface.co/datasets/VISION-Workshop

VISION Datasets — what makes it different

Three design differentiators.

- ▶ **Versatility.** Annotation masks on *all* train / val / test splits — supports unsupervised, weakly-, semi-, and fully-supervised algorithms.
- ▶ **Instance-segmentation labels** (not just semantic): distinguishes multiple defects *of the same type* in a single image.
- ▶ **Scale and diversity.** Largest and one of the most diverse vision-based inspection benchmarks to date.

Curation pipeline.

- ▶ 1,800+ Roboflow datasets screened down to the top 14.
- ▶ Defect-level deduplication + connected-component analysis $\Rightarrow$ train / val / test split with minimal leakage.
- ▶ 3,200 annotation hours; ambiguous boundaries arbitrated to consensus.

Comparison with prior inspection benchmarks

Dataset	Images	Masks	Types
AITEX	205	105	12
BeanTech	1,799	290	3
KolektorSDD	399	52	1
KolektorSDD2	3,335	356	1
MTD	1,344	392	5
MVTec AD	5,354	1,258	69
VISION V1	18,422	9,837	44
VISION V2	18,422	38,272	44

V1 is released; V2 adds full instance-segmentation annotations (released with the subsequent challenge).

Thank you — Questions?



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Questions?

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